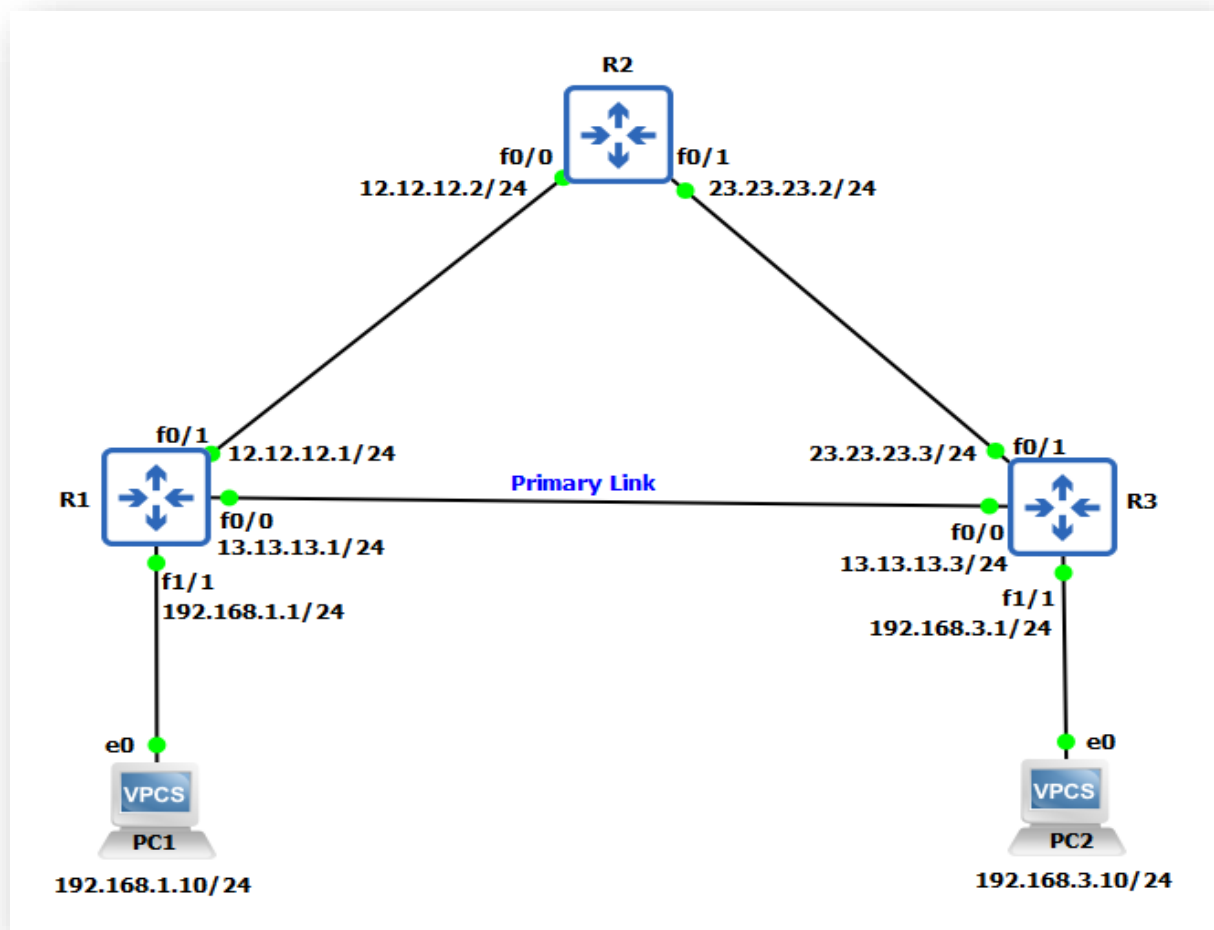


Default Floating Routing



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** and device access password as **cisco** (Refer Lab-1)
2. Configure static routes (next-hop) on R2 for R1 and R3 LAN network.
3. Configure default route on R1 & R3 for primary link.
4. Configure default floating route on R1 for R3's LAN network with Administrative Distance of 15
5. Configure default floating route on R3 for R1's LAN network with Administrative Distance of 15
6. PC1 & PC2 should ping each other.

R1 router configuration:

```

R1#config t
R1(config)#ip route 0.0.0.0 0.0.0.0 13.13.13.3
R1(config)#ip route 0.0.0.0 0.0.0.0 12.12.12.2 15
R1(config)#exit
R1#

```

R2 router configuration:

```
R2#config t
R2(config)#ip route 192.168.1.0 255.255.255.0 12.12.12.1
R2(config)#ip route 192.168.3.0 255.255.255.0 23.23.23.3
R2(config)#exit
R2#
```

R3 router configuration:

```
R3#config t
R3(config)#ip route 0.0.0.0 0.0.0.0 13.13.13.1
R3(config)#ip route 0.0.0.0 0.0.0.0 23.23.23.2 15
R3(config)#exit
R3#
```

✓ Verification & Testing:**R1#sh run | section ip route**

```
ip route 0.0.0.0 0.0.0.0 13.13.13.3
ip route 0.0.0.0 0.0.0.0 12.12.12.2 15
R1#
```

R1#sh ip route static | begin Gateway

```
Gateway of last resort is 13.13.13.3 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [1/0] via 13.13.13.3
R1#
```

R3#sh run | section ip route

```
ip route 0.0.0.0 0.0.0.0 13.13.13.1
ip route 0.0.0.0 0.0.0.0 23.23.23.2 15
R3#
```

R3#sh ip route static | begin Gateway

```
Gateway of last resort is 13.13.13.1 to network 0.0.0.0
```

```
S* 0.0.0.0/0 [1/0] via 13.13.13.1
R3#
```

R1#sh ip route 192.168.3.10

```
% Network not in table
```

R1#ping 192.168.3.10

```
Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:
```

```
..!!!!
```

```
Success rate is 80 percent (4/5), round-trip min/avg/max = 52/189/564 ms
```

R1#traceroute 192.168.3.10

```
Type escape sequence to abort.
```

Tracing the route to 192.168.3.10
VRF info: (vrf in name/id, vrf out name/id)
1 13.13.13.3 56 msec 56 msec 52 msec
2 192.168.3.10 56 msec 56 msec 56 msec
R1#

R3#sh ip route 192.168.1.10

% Network not in table

R3#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/56/60 ms

R3#

R3#traceroute 192.168.1.10

Type escape sequence to abort.

Tracing the route to 192.168.1.10

VRF info: (vrf in name/id, vrf out name/id)

1 13.13.13.1 60 msec 56 msec 56 msec
2 192.168.1.10 56 msec 56 msec 52 msec

R3#

PC1> trace 192.168.3.10

trace to 192.168.3.10, 8 hops max, press Ctrl+C to stop

1 192.168.1.1 9.153 ms 9.155 ms 10.135 ms
2 13.13.13.3 30.121 ms 30.262 ms 32.263 ms
3 *192.168.3.10 42.271 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> trace 192.168.1.10

trace to 192.168.1.10, 8 hops max, press Ctrl+C to stop

1 192.168.3.1 4.292 ms 9.265 ms 9.326 ms
2 13.13.13.1 31.220 ms 31.225 ms 31.240 ms
3 *192.168.1.10 42.298 ms (ICMP type:3, code:3, Destination port unreachable)

Simulate the primary link failure by shutting down interface fa0/0 on R1 & R3 router.

R1 router configuration:

R1#config t
R1(config)#int fa0/0
R1(config-if)#shut
R1(config-if)#exit
R1(config)#exit
R1#

***Oct 10 19:36:30.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down**

R3 router configuration:

```
R3#config t
R3(config)#int fa0/0
R3(config-if)#shut
R3(config-if)#exit
R3(config)#exit
R3#
```

***Oct 10 19:38:12.721: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down**

✓ Verification & Testing:**R1#sh ip int brief | exclude unassigned**

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	13.13.13.1	YES	manual	administratively down	down
FastEthernet0/1	12.12.12.1	YES	manual	up	up
FastEthernet1/1	192.168.1.1	YES	manual	up	up

R1#

R3#sh ip int brief | exclude unassigned

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	13.13.13.3	YES	manual	administratively down	down
FastEthernet0/1	23.23.23.3	YES	manual	up	up
FastEthernet1/1	192.168.3.1	YES	manual	up	up

R3#

R1#sh ip route static | begin Gateway

Gateway of last resort is 12.12.12.2 to network 0.0.0.0

S* 0.0.0.0/0 [15/0] via 12.12.12.2

R1#

R1#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

..!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 80/202/564 ms

R1#

R1#traceroute 192.168.3.10

Type escape sequence to abort.

Tracing the route to 192.168.3.10

VRF info: (vrf in name/id, vrf out name/id)

```
 1 12.12.12.2 56 msec 60 msec 64 msec
 2 23.23.23.3 80 msec 104 msec 84 msec
 3 192.168.3.10 80 msec 80 msec 80 msec
```

R1#

R3#sh ip route static | begin Gateway

Gateway of last resort is 23.23.23.2 to network 0.0.0.0

S* 0.0.0.0/0 [15/0] via 23.23.23.2

R3#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

.!!!!

Success rate is 80 percent (4/5), round-trip min/avg/max = 80/206/576 ms

R3#

R3#traceroute 192.168.1.10

Type escape sequence to abort.

Tracing the route to 192.168.1.10

VRF info: (vrf in name/id, vrf out name/id)

- 1 23.23.23.2 56 msec 56 msec 56 msec
- 2 12.12.12.1 56 msec 52 msec 84 msec
- 3 192.168.1.10 80 msec 80 msec 84 msec

R3#

PC1> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=61 time=61.080 ms

84 bytes from 192.168.3.10 icmp_seq=2 ttl=61 time=65.114 ms

84 bytes from 192.168.3.10 icmp_seq=3 ttl=61 time=62.218 ms

84 bytes from 192.168.3.10 icmp_seq=4 ttl=61 time=55.261 ms

84 bytes from 192.168.3.10 icmp_seq=5 ttl=61 time=55.432 ms

PC1> trace 192.168.3.10

trace to 192.168.3.10, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.1 3.154 ms 10.171 ms 10.141 ms
- 2 12.12.12.2 31.198 ms 32.286 ms 32.176 ms
- 3 23.23.23.3 54.169 ms 51.166 ms 51.171 ms
- 4 *192.168.3.10 64.177 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=61 time=56.149 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=61 time=62.156 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=61 time=61.210 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=61 time=57.223 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=61 time=63.240 ms

PC2> trace 192.168.1.10

trace to 192.168.1.10, 8 hops max, press Ctrl+C to stop

- 1 192.168.3.1 5.163 ms 9.292 ms 9.279 ms
- 2 23.23.23.2 31.161 ms 31.192 ms 31.225 ms
- 3 12.12.12.1 53.135 ms 51.213 ms 53.204 ms
- 4 *192.168.1.10 63.213 ms (ICMP type:3, code:3, Destination port unreachable)

Important Commands:

```
sh ip route  
sh ip route static  
sh ip route static | begin Gateway  
sh ip int brief | exclude unassigned
```

Ratan